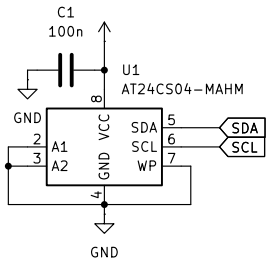
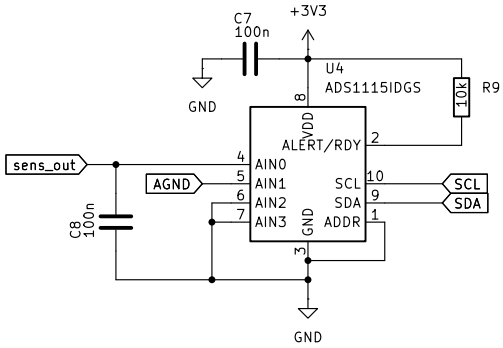
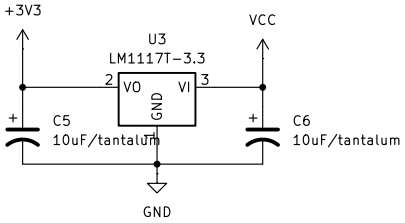
MCP9808 digital temperature sensor, ±0.25 °C typ, 0.0625 °C resolution,
I2C address 0x18 (A0, A1, A2 = GND). VDD from +3V3 with 100n at the pin.
Purpose: track the detector die temperature, not ambient. Log detectors
drift roughly ±0.3 dB over −40 to +85 °C; the firmware reads this sensor
(MEAS:TEMP?) so each head's drift can be characterised and compensated.
Place as close to the AD8307 as possible, same side of the board, on the
same ground pour, with no slot or thermal relief between them.



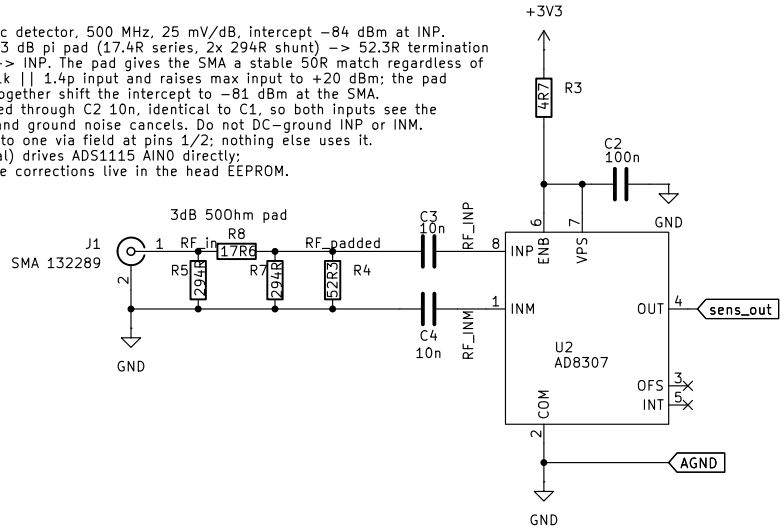
ADS1115 − 16−bit delta−sigma ADC, I2C address 0x48 (ADDR=GND)
Measures AD8307 OUT differentially against detector COM (AIN0−AIN1),
PGA ±4.096 V, continuous mode 128 SPS, set by firmware on hot−plug.
C_filt 100 nF with the AD8307's 12.5 kΩ output impedance: ~130 Hz video filter.
AIN1 routed to AD8307 COM pin, not to a nearby GND via.
ALERT/RDY unused, pulled up.



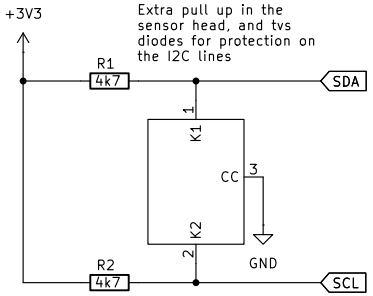
The VCC on the cable from
the controller is at 8V or
thereabouts, for the AD8307
it is best to control down to
3V3, since the I2C from the
controller sits at 3V3.



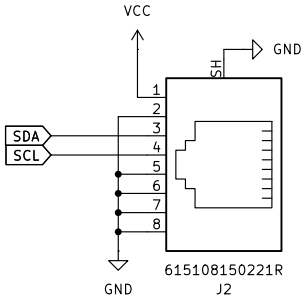
AD8307 logarithmic detector, 500 MHz, 25 mV/dB, intercept −84 dBm at INP.
RF path: SMA −> 3 dB pi pad (17.4R series, 2x 294R shunt) −> 52.3R termination
−> C1 10n C0G −> INP. The pad gives the SMA a stable 50R match regardless of
the detector's ~1.1k || 1.4p input and raises max input to +20 dBm; the pad
plus termination together shift the intercept to −81 dBm at the SMA.
INM is AC−grounded through C2 10n, identical to C1, so both inputs see the
same impedance and ground noise cancels. Do not DC−ground INP or INM.
R4 and C2 return to one via field at pins 1/2; nothing else uses it.
OUT (12.5k internal) drives ADS1115 AIN0 directly;
intercept and slope corrections live in the head EEPROM.



Extra pull up in the
sensor head, and tvs
diodes for protection on
the I2C lines



The cable from the controller
is connected through an RJ−45
connector − a model which is low
profile with respect to the PCB.
VCC is 8V, I2C sits at 3V3.



Device	I2C address	Function
ADS1115	0x48	AD converter
MCP9808	0x18	temperature sensor
AT24CS04	0x50	EEPROM lower 256 bytes
AT24CS04	0x51	EEPROM upper 256 bytes
AT24CS04	0x58	128 bit serial no.

Erik Ø. Sørensen, LB3TH

Sheet: /
File: AD8307.kicad_sch

Title: RF Power Meter, Sensor Head: AD8307 Logarithmic Detector

Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/1